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Paleomagnetic studies on the late Ediacaran – Early Cambrian Puncoviscana and the late Cambrian Campanario formations, NW Argentina: New paleogeographic constraints for the Pampia terrane

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ABSTRACT

A magnetofabric and paleomagnetic study was carried out in the Late Ediacaran – Early Cambrian Puncoviscana and the early Late Cambrian Campanario formations, exposed in Santa Victoria Oeste, in northwestern Argentina. Ten sites (93 samples) were located in tuffs and volcanic sills interbedded in the Puncoviscana Formation, one of which had been dated at 537 ± 0.9 Ma. On the other hand, 42 samples were collected at three sites from red to purple sandstones of the Campanario Formation. The analysis of anisotropy of magnetic susceptibility (AMS) allowed to recognize a pre-Andean fabric in both formations indicating that it is previous to the Andean cycle. In the paleomagnetic study reliable magnetic components were isolated in only two sites of the Puncoviscana formation whose virtual geomagnetic poles (VGPs) are close to but do not coincide with the apparent polar wander path of Río de la Plata craton – Gondwana for the late Neoproterozoic – Cambrian. A new paleomagnetic pole (18.2°S, 358.8°W, K: 27.9; A95: 3.9) was computed for the Campanario formation which is close to but does not coincide with those obtained in other locations for this unit and were considered anomalous respect to the expected position in the Gondwana path. The pole presented here is closer to the apparent polar wander path of Gondwana for the Late Cambrian-Early Ordovician suggesting two possible interpretations; the presence of different amounts of Andean tectonic rotations between different sampling locations of the Campanario Formation or the recording of a rapid Pampia dextral displacement along the Rio de la Plata craton margin in Cambrian times.

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1. Introduction

Paleomagnetic studies are an important tool in the reconstruction of the global paleogeographic evolution (e.g. Van der Voo, 1993), and allow to solve controversies between opposing tectonic models for a region (e.g. Rapalini, 2005). The construction of the South American Gondwana margin has been a complex process, whose details are not yet known with precision and have resulted in a variety of different tectonic models (Cawood, 2005; Rapela et al., 1998, 2007, 2016; Ramos, 2008; Ramos et al., 2010 and many others). In the last two decades some models have proposed a latest Proterozoic to Paleozoic accretion of various terranes or crustal blocks to the Gondwana cratonic margin represented at these latitudes by the Rio de la Plata craton of Paleoproterozoic age (see Rapela et al., 2007). The main terranes located to the west of the craton (see Fig. 1) are Pampia, also called Pampean terrane (Ramos et al., 2010; Rapela et al., 2016), Arequipa-Antofalla block (Ramos, 2008), Cuyania terrane (Ramos, 2004) and Chilenia (Ramos et al., 1984; Davis et al., 1999) (Fig. 1). Several authors have proposed partially different configurations as well as ages and modes of accretion for these blocks. Since the original proposal by Ramos (1988) and Escayola et al. (1996) of a frontal collision between Pampia and the Río de la Plata craton in the Neoproterozoic, several models have been published with different scenarios and processes for the relationship of these two blocks. For instance, Rapela et al. (2007, 2016) proposed a strike-slip displacement and accretion of Pampia along the western (present coordinates) margin of the Río

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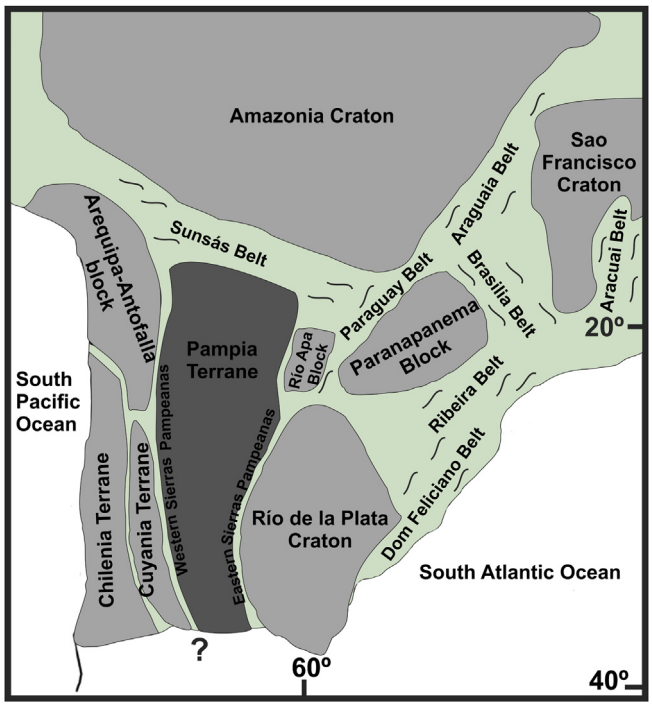


Fig. 1. The Pampia terrane and its relationship with adjacent cratons and Paleozoic terranes in southern South America. Modified from Rapalini (2005), Ramos et al. (2010) and Rapela et al. (2016).

de la Plata craton, prior to the Late Cambrian. According to this model, Pampia is a fragment of a continental shelf of the Kalahari craton that was dismembered and translated along the Gondwana margin up to its present-day position during the final stages of Gondwana assembly. Spagnuolo et al. (2008, 2012), based on paleomagnetic studies on the late middle to early late Cambrian Mesón Group proposed a model compatible with that proposal, suggesting a dextral displacement of Pampia during the Cambrian from the present southern Kalahari edge to a final position close to Rio de la Plata craton before the Early Ordovician. Since the main anomaly in the pole position for the Meson Group can be attributed to a clockwise rotation about a vertical axis, Spagnuolo et al. (2008) suggested an alternative interpretation that involved local tectonic rotations associated with the Andean uplift and the Bolivian Orocline (eg Arriagada et al., 2006; Prezzi et al., 2004, 2014; Somoza et al., 1996; Somoza and Tomlinson, 2002). This alternative interpretation was considered less likely later on by Spagnuolo et al. (2012) as similar paleomagnetic directions were subsequently obtained for the Mesón Group in a second locality over thirty kilometers apart from the first one and the fact that paleomagnetic directions from Early Ordovician sandstones in the same locality indicated a less significant paleomagnetic anomaly (Spagnuolo et al., 2012). To unravel the kinematic history of Pampia and its relationship to the Río de la Plata craton is of significance to reconstruct the final processes of Gondwana assembly and the crustal structure of the southern part of South America. For these reasons a new paleomagnetic study was carried out in the Campanario formation of the Mesón Group, in a third locality, over 50 km away from the previous two localities studied by Spagnuolo et al. (2008, 2012). This was complemented with the first paleomagnetic study on volcanic and pyroclastic levels of the Puncoviscana formation (late Ediacaran – early Cambrian) which lies under the Mesón Group deposits.

2. Geological setting

During the Late Neoproterozoic – Early Paleozoic the proto-Pacific margin of Gondwana was affected by a series of extensional basins ranging from southern South America through Antarctica to Australia (e.g. Cawood, 2005).

In northwestern Argentina, this episode is interpreted to be represented by the Puncoviscana formation. It consists of a complex sedimentary succession with alternating siliciclastic rocks, commonly graded, tuffs, siltstones (Escayola et al., 2011) with a low to very low metamorphic grade (Do Campo and Nieto, 2003). The siliciclastic rocks have been interpreted as representing mainly turbiditic flows (Jezek et al., 1985) and it has a volcanic origin, with a significant participation of sanidine and numerous glass shreds.

The Puncoviscana Formation shows a complex structuration with several orders of tight folding, commonly in chevron style (Willner et al., 1987; Piñan-Llamas and Simpson, 2006). Structures of this unit are truncated by a spectacular unconformity assigned to the Tilcaric tectonic phase (Turner and Méndez, 1975), which has been dated as early Cambrian (ca. 525 Ma).

The age of the Puncoviscana formation is generally inferred to be late Ediacaran to early Cambrian, based on the presence of the gender *Oldhamia* (Buatois and Mángano, 2003) and its position immediately below the Tilcaric unconformity that separates it from the overlying middle to late Cambrian Mesón Group. Age also can be restricted from that of syn and post tectonic intrusive bodies. The Cañani pluton has been dated as 523 ± 5 Ma and the Cerro Negro orthogneiss in the southern domain of the Arequipa-Antofalla block as 523 ± 1 Ma (Escayola et al., 2011). The Tastil granodiorite has provided ages of 534 ± 7 Ma (Matteini et al., 2008; Hauser et al., 2010) and 517 ± 2 Ma (Hongn et al., 2010). The Cañani and Tastil plutons were involved in the Tilcaric orogenic phase, which ended prior to the deposition of the Mesón Group sandstones in the middle to late Cambrian.

Recent advances in a more precise age determination for the Puncoviscana formation have been provided through U-Pb dating of detrital zircons (Zimmermann, 2005; Adams et al., 2008, 2010), which yielded maximum sedimentation age suggesting that at least the upper levels of the formation were deposited in the early Cambrian. Recently, Escayola et al. (2011) obtained an U-Pb (SHRIMP) age on zircons from an acidic tuff at the type locality of this formation (Puncoviscana town) of 537 ± 1 Ma.

The origin of the Puncoviscana formation have been assigned to a variety of different environments and processes, i.e. foreland basin (Kraemer et al., 1995; Keppie and Bahlburg, 1999; Zimmermann, 2005), passive margin (Jezek et al., 1985; Adams et al., 2008, 2010; Schwartz et al., 2008) or rift basin (Omarini et al., 1999; Ramos, 2008; Collo et al., 2009). Escayola et al. (2011) interpreted the Puncoviscana basin as developed in two stages: i) arc-trench gap during the Ediacaran late Early Cambrian (>530 Ma), ii) syncolisional foreland basin (Tilcaric Orogeny) (<530 Ma), formed as a result of the thrust of the Pampean arc over the Arequipa-Antofalla block. For the overlying Mesón Group there is a greater consensus on a shallow platform environment.

The Mesón Group consists of silicified sandstones (orthoquartzites), sandstones and conglomerates, with minor intercalations of shales (Turner, 1964). Its thickness varies from tens of meters in marginal parts of the basin up to several hundred meters. It presents no metamorphism and mild deformation. Keidel (1943) proposed to divide the Group into three formations: Lizoite, Campanario and Chalhualmayoc, respectively, from base to top. The Campanario Formation, studied in this work, consists of green and purple sandstone, characterized by the presence of Skolithus tubes with intercalated shale levels.

Adams et al. (2008), based on detrital zircons in the Mesón

Group assigned a maximum depositional age of 502 Ma. The paleontological evidence constrain the age of the whole Group to the Middle to Late Cambrian (Sanchez and Salfity, 1999; Aceñolaza, 2003; Aceñolaza and Aceñolaza, 2005). The Mesón Group is covered by the sandstones of the Santa Rosita Formation through a mild unconformity (Iruyica unconformity, Astini et al., 2008), close to the Cambro-Ordovician boundary.

2.1. Geology of the study area

In its type locality the Puncoviscana Formation shows excellent outcrops along the northern and southern slopes of the Acoyte river valley (Fig. 2a). These outcrops consist of turbidite sequences with layers of sandstones over 20 cm thick and intercalations of fine tuffs levels of about 10 cm thick, which become more frequent towards the top of the local succession. Both rock types are interleaved with 10 cm thick layers of shale with parallel lamination. The succession was affected by folding and multiple reverse faults that do not allow to construct a continuous stratigraphic profile beyond a few tens of meters thick. Numerous quartz veins are found affecting the entire sequence.

The stratification is affected by the presence of some basaltic dikes or sills of one to two meters thick, usually disposed concordantly with bedding of the surrounding banks. The intrusive character of the basaltic levels is clearly exposed at P1 site, where sediments in contact with the upper border of the sill are affected by contact metamorphism (Fig. 3b). At site P3 the basaltic sill is intruded by a subvertical dacitic dike corresponding to a subsequent volcanic event.

Outcrops of the Puncoviscana formation in the area also show an overimposed important tectonic shortening product of the Andean orogeny and expressed through a succession of folding and thrust faults that generated several tectonic slices of small dimensions with systematic vergence towards the NE. On a smaller scale, there are several short wave length folds (<1 m), some of which developed chevron geometries (Fig. 2d).

The discordant contact with the overlying Meson Group is visible on the western boundary of the study area, where the sandstones of Lizoite formation cover the Puncoviscana formation. The Meson Group appears like a homoclinal succession affected by reverse faulting. The passage from the yellowish to white sandstones of the Lizoite Formation to the purple and greenish sandstones of the Campanario formation is very well exposed along the margins of the valley of the Acoyte river. These exposures consist in intercalation of 30–40 cm thick sandstones banks with shales levels of no more than 15 cm thick.

3. Sampling and methodology

A magnetofabric and paleomagnetic study was carried out on 93 cores (161 specimens) collected at 10 sites (P1, P2, P3, P4, P5, P6, P7, P8, P9 and P10) located in volcanic and volcaniclastic rocks of the Puncoviscana formation in the Santa Victoria Oeste locality (Fig. 2a). Local attitudes were variable with west-dippings from 4° to 70°. At each site generally 10 independently oriented cores were collected with a portable drill. This sampling was complemented with 42 independently oriented cores collected at three sites on the Campanario formation (M2, M3 and M4) and 10 independently oriented cores at a single site (M1) on an undated acidic intrusive body in the Lizoite formation. Sampling of the Campanario formation spanned a stratigraphic interval exceeding 30 m. The horizontal distance between M2 and M3 is approximately 60 m and between M3 and M4 about 25 m. The sampled banks are exposed as an approximately homoclinal succession with dips ranging between 22° and 42° towards the West and strike azimuths ranging between 175° and 200°. Nine, twenty-seven and six independently oriented cores were collected from M2, M3 and M4 sites, respectively. The intrusive body M1 is located ca. 700 m from the Campanario Formation sites (Fig. 2a). It consists in a concordant sill of an acidic volcanic rock about 7 m thick that intrudes the underlying

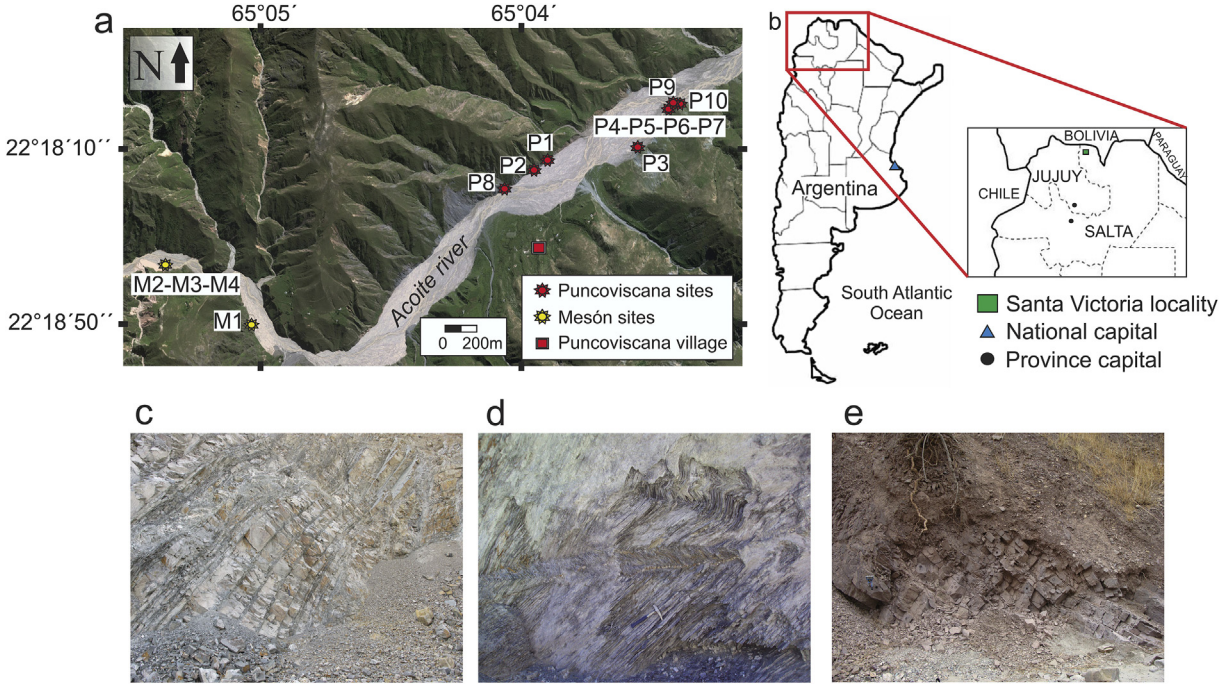


Fig. 2. a) Sites sampled at the Puncoviscana and Campanario formations in the locality of Santa Victoria Oeste; b) location of the study area; c) view of an outcrop of the Puncoviscana formation in the study area. Note the important difference in thickness between the turbidite sequences banks and the tuff levels on top; d) chevron-type deformation observed in the Puncoviscana formation; e) outcrop of the Campanario formation in the study area.

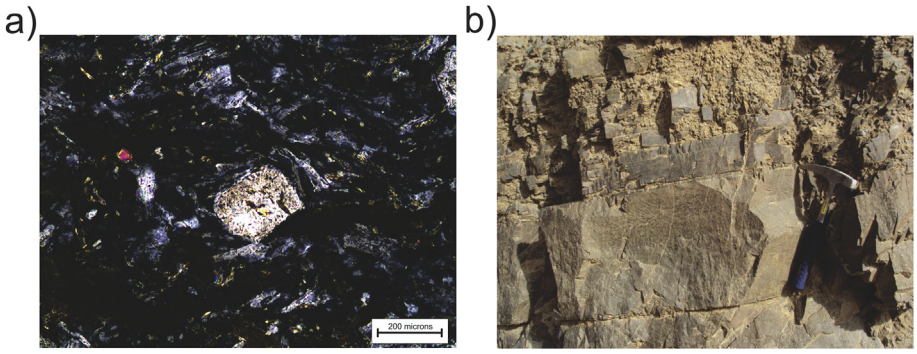


Fig. 3. a) Sigma (σ) type structure at site P3. Note the flow texture of plagioclase microlites surrounding the phenocryst; b) Contact metamorphism generated at the contact between a basalt sill (site P1) and the overlying shale of the Puncoviscana Formation.

Lizoite Formation. Other similar sills are exposed in the study area. All cores were oriented with magnetic and sun compasses, and inclinometer. Differences between both systems were never higher than a few degrees and both measurements were generally averaged. All cores were afterwards sliced into standard-size specimens of 2.54 cm in diameter and 2.2 cm high in the Laboratory of Paleomagnetism Daniel A. Valencio of IGEBA (University of Buenos Aires).

Measurements of the anisotropy of magnetic susceptibility (AMS) were done with an MKF1-A susceptibilimeter (AGICO Inc). Measurements of the intensity and direction of the natural remanent magnetization were done with a JR6 (AGICO Inc) spinner magnetometer. Stepwise demagnetization was accomplished with a tumbler LD-A (AGICO Inc) alternating field demagnetizer and a single chamber ASC-48TD furnace.

Isothermal remanence acquisition and backfield curves (up to 2 T) were obtained from representative samples with an ASC pulse magnetizer. Thermomagnetic curves (k vs T) were obtained for representative samples of the Puncoviscana and Campanario formations. The procedure was carried out using the MFK1-A susceptibilimeter (AGICO) with CS-L and CS3 devices. Low temperatures curves were obtained by cooling a sample with liquid nitrogen up to around −188 °C and bulk susceptibility monitoring during heating up to room temperature. High temperatures curves were performed in an inert atmosphere by continuous injection of Ar from ambient temperature to 680 °C and cooling back to room temperature.

4. AMS study

4.1. Puncoviscana Formation

Site P1 shows very different characteristics with respect to the other sites, in the bulk susceptibility, the distribution of the anisotropy axes, the anisotropy degree and the shape of the ellipsoid (Table 1).

On the other hand, P2–P10 (Fig. 4) show a pre-Andean magnetic fabric, which is evidenced by the much better directional consistency of the AMS axes after bedding correction. K3 axis is subvertical or dipping steeply to the west in paleohorizontal coordinates. All these sites show oblate ellipsoids (T > 0), bulk susceptibility values between 10^{-4} and 3.10^{-2} SI and low to moderate anisotropy degrees (up to 21%).

By contrast, P1 presents *in situ* K3 axes with moderate dips in the NW direction that become subhorizontal after bedding untilting. Microscopic analysis suggests a primary volcanic fabric and lack of metamorphism or severe internal deformation. Samples from this site show both oblate and prolate ellipsoids, higher bulk susceptibilities ($10^{-2} < k < 6 \times 10^{-2}$ SI) and low anisotropy degrees (<5%).

While the basaltic sill of site P1 has a high magnetic susceptibility, due to a significant presence of magnetite, as verified by microscopic observation (see Section 7), thinner basaltic sills sampled at sites P3 and P9 presented a higher degree of alteration, with significant oxidation of magnetite to hematite. This is probably the cause of the much lower bulk susceptibility measured at these sites. In all three cases, however, the anisotropy degree is low.

In the case of the volcaniclastic rocks (remaining sites), no significant differences among them were observed. The presence of magnetite at these sites is confirmed by rock-magnetic studies (see below). The lack of correlation between K and P suggests that the degree of anisotropy is related to the degree of deformation. The significant flattening shown by the magnetic fabric of the volcaniclastic and some basaltic sites correlates with abundant microscopic evidence of deformation (Fig. 3a). The pre-Andean character of the magnetofabric in these rocks suggests that it may be associated with deformation and low grade metamorphism of the Tilcaric phase in the Early Cambrian.

4.2. Campanario Formation

M2, M3 and M4 sites at the Campanario formation show a pretectonic magnetic fabric, observed from the subvertical position of K3 axes after bedding correction and the coincidence of the magnetic foliation (K1–K2) with the bedding plane (Fig. 4). The fabric is oblate (T > 0) and shows a correlation between the anisotropy degree and the bulk susceptibility, typical of fabrics governed by ferromagnetic minerals (Lopez de Luchi et al., 2010 and references therein). The anisotropy is relatively low as it is less than 6%, suggesting that degree of compaction is probably not high. Thin sections observations confirmed lack of dynamic metamorphism in these rocks. The fabric pattern is typical of a depositional or compactional origin during diagenesis, with no defined magnetic lineation. Bulk susceptibility values are between 10^{-5} and 2×10^{-4} SI.

The volcanic sill sampled at M1 has a distribution of anisotropy axes that precludes evaluating whether the magnetic fabrics is pre- or postectonic. In situ distribution of AMS axes show a well defined NNE-SSW subvertical foliation plane, although a few samples seem to form a second group with a different distribution. The fabric pattern resembles more one of a vertical dike than a concordant sill. Bulk susceptibility ranges from 1.8 to 3.2×10^{-4} SI. Anisotropy degree is low (<8%) suggesting a magmatic fabric.

Table 1
Mean site bedding corrected AMS parameters: *k*1, *k*2 and *k*3 correspond to the maximum, intermediate and minimum axes of the ellipsoid, respectively. Their directions (declination/inclination) are presented with the respective 95% confidence ovals (maximum and minimum semiaxes are quoted in italics between brackets). BP is the bedding attitude (strike/dip) for each site, with dip measured respect to the horizontal following the right-hand rule. Kmean stands for the mean site bulk susceptibility in SI units. Pj is the corrected anisotropy factor (Jelinek, 1978). T is Jelinek's (1981) shape parameter.

Site	Lithology	BP (°)	k1 (°)	k2 (°)	k3 (°)	Kmean	Pj	T
P1	Basalt	142/52	177/41 (*11/8*)	43/38 (*12/10*)	291/25 (*12/8*)	3.68E-02	1.034	−0.051
P2	Tuff	175/39	87/10 (*19/1*)	356/5 (*19/2*)	240/79 (*2/2*)	1.93E-03	1.133	0.61
P3	Basalt	27/7	58/14 (*15/2*)	149/4 (*15/3*)	255/76 (*4/3*)	1.52E-04	1.047	0.603
P4	Tuff	345/25	89/24 (*5/2*)	358/3 (*5/3*)	261/65 (*5/2*)	1.31E-02	1.086	0.225
P5	Volcaniclastic	339/37	100/14 (*4/2*)	9/4 (*4/2*)	265/76 (*3/2*)	7.00E-03	1.190	0.432
P6	Tuff	339/37	113/1 (*4/3*)	23/14 (*3/2*)	206/76 (*4/2*)	4.36E-03	1.180	0.448
P7	Tuff	339/37	106/9 (*5/2*)	14/12 (*4/2*)	234/75 (*4/2*)	3.31E-03	1.204	0.471
P8	Volcaniclastic	175/39	118/6 (*18/2*)	208/5 (*11/4*)	337/83 (*16/4*)	2.70E-04	1.062	0.391
P9	Basalt	342/38	76/26 (*19/9*)	170/7 (*19/7*)	275/63 (*10/8*)	2.12E-04	1.055	0.659
P10	Volcaniclastic	3/57	100/5 (*5/2*)	10/1 (*4/3*)	269/85 (*5/3*)	6.92E-03	1.171	0.293
M1	Volcanic sill	175/34	19/3 (*20/7*)	283/66 (*20/11*)	110/24 (*12/5*)	2.41E-04	1.045	0.338
M2	Sandstones	181/39	358/2 (*20/5*)	268/9 (*21/12*)	103/81 (*12/6*)	8.59E-05	1.020	0.352
M3	Sandstones	189/26	189/2 (*29/4*)	280/9 (*29/7*)	85/81 (*7/4*)	1.24E-04	1.037	0.786
M4	Sandstones	174/35	23/9 (*12/3*)	113/3 (*12/9*)	223/81 (*9/2*)	1.35E-04	1.024	0.712

5. Paleomagnetic results

For the paleomagnetic analysis of both formations a pilot study was first carried out consisting in a detailed stepwise demagnetization procedure of four specimens per site, two through alternating magnetic fields (AF), and the others by high temperatures (Butler, 1992). Results from the pilot study were used to choose the best demagnetization procedure for the remaining samples of each site.

AF demagnetization was selected for sites P1 and P5 whereas for the remaining sites of the Puncoviscana formation, all sites of the Campanario formation and the single site of the intrusive body (M1) high temperature demagnetization was considered more effective. AF demagnetization steps generally used were: 3, 6, 9, 12, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80 mT, whereas high-temperature demagnetization proceeded at the following temperature steps: 150, 200, 250, 300, 350, 400, 450, 500, 550, 580 and 600 °C for P6, P7, P9 and P10, while for P2 and P8 they were: 150, 300, 400, 500, 550, 600, 625, 650, 675 °C. For all of the specimens of the Campanario formation steps were slightly different, at 100, 150, 200, 250, 300, 350, 400, 450, 500, 530, 560, 590, 615, 635, 650 and 670 °C. Each specimen was analyzed individually and the magnetic components determined by principal component analysis (PCA, Kirschvink, 1980).

Sites P3 and P4 showed a very unstable magnetic remanence during the pilot study for both alternating magnetic fields (AF) and high temperatures methods. At these sites no stable components of remanent magnetization could be isolated.

5.1. Puncoviscana formation

Sites P2, P5, P6, P7, P8, P9 and P10 carried a magnetic component of low unblocking temperatures and/or coercive forces that was labelled as component “a” and usually isolated at fields under 20 mT and temperatures lower than 350 °C (Fig. 5). This component was determined with moderate accuracy in many specimens of different sites. Maximum angular deviations (MAD) under 12° were accepted in this case.

Component “a” has a relatively poor *in situ* clustering (Dec: 359.1°; Inc: −21.3°; n: 7; k: 19.1; α95: 14.2°), that worsens after bedding correction (Dec: 345.1°; Inc: −18.8°; n: 7; k: 14.7; α95: 16.3°), suggesting that it is a post-tectonic secondary component. Its proximity to the present-day geocentric axial dipole (GAD) direction for the study area also points to a secondary overprint (Fig. 6).

A second component with higher unlocking temperatures and/or coercive forces, labelled component “b” was identified at site P1, named component “c” for site “P2” and called component “d” for sites P5, P6, P7 and P10 (Fig. 5). At sites P1 and P2 the components showed a very good internal directional consistency, but with very different mean directions (Table 2).

Component “b” at P1 (Fig. 7), was mainly determined by AF demagnetization in ten specimens, plus two other by thermal treatment. They show maximum unblocking temperatures between 500 and 550 °C and destructive fields between 30 and 40 mT. Each component was determined with MAD <12°. The average direction obtained for P1 *in situ* is: Dec: 40.9°; Inc: −38.4°; n: 12; k: 66.7; α95: 5.4°, while after bedding correction is Dec: 43.1°; Inc: 13.0° (Table 2).

At P2 demagnetization was carried out by thermal methods with an unblocking temperature between 600 and 625 °C. Except for a single specimen, in all others component “c” was determined with MAD under 10°. The average direction obtained for P2 *in situ* is: Dec: 29.5°; Inc: 50.1°; n: 11; k: 57.1; α95: 6.1°, while after bedding correction is Dec: 334.3°; Inc: 55.5°.

For the remaining sites in which a high temperature/coercivity component (“d”) could be isolated, (P5, P6, P7 and P10) a moderate to poor intra-site and between-sites directional consistency were observed (Fig. 7a). Site P5 was mainly demagnetized by AF with relatively low precision, involving specimens with MAD below 15°. The two specimens thermally demagnetized showed unblocking temperatures between 500 and 550 °C. P6, P7 and P10 sites were mainly thermally demagnetized with unblocking temperatures ranging between 550 and 580 °C. In all specimens component “d” was determined with MAD under 15°. The average direction obtained from the specimens of these four sites *in situ* is Dec: 352.7°; Inc: −31.0°; n: 14; k: 7.2; α95: 16.0°, while the direction obtained after bedding correction shows an improvement in clustering: Dec: 328.1°; Inc: −25.7°; n: 14; k: 8.8; α95: 14.2°. The mean direction *in situ* obtained from the site mean values is: Dec: 346.8°; Inc: −30.6°; N:4; k: 12.6; α95:27°, and after bedding correction is Dec: 324.6°; Inc: −23°; k: 23.8; α95: 19.2°. The improvement in the statistical parameters after bedding correction suggests that component “d” is pre-Andean, however application of the direction-correction tilt test of Enkin (2003) to the site mean directions is inconclusive.

Sites P8 and P9 presented inconsistent remanence directionsat higher temperatures and magnetic fields, and no high temperature/coercivity component could be determined.

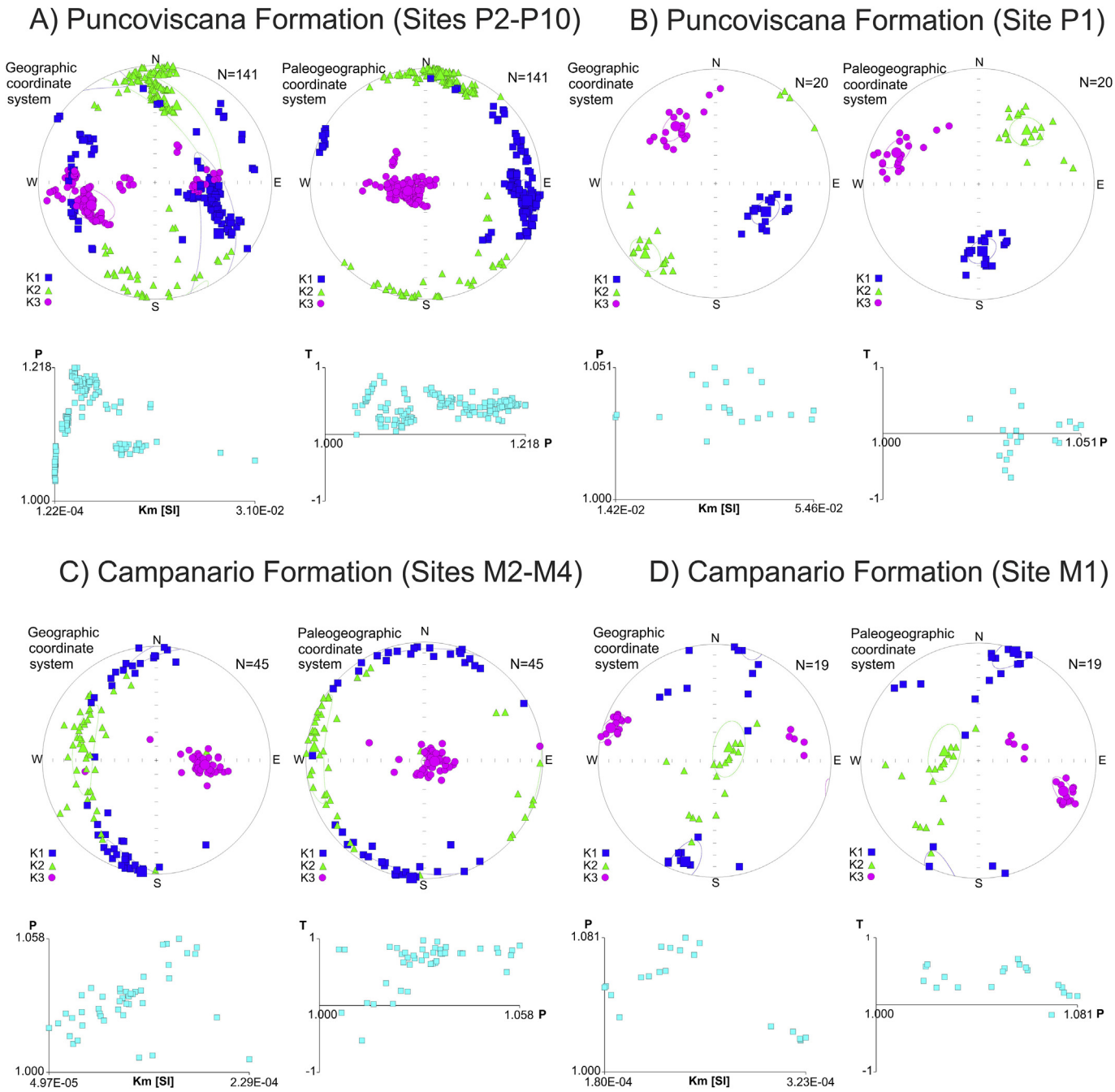


Fig. 4. Stereographic projections of the principal axes of the ellipsoid of anisotropy of magnetic susceptibility (K1, K2 and K3) for the Puncoviscana and Campanario Formations samples and their respective Pj vs. Km and T vs. Pj plots. A) stereographic projection and plots Pj vs. Km and T vs. Pj from sites P2 to P10 of the Puncoviscana formation; B) Idem for P1 site; C) Idem for M2 to M4 sites of the Campanario formation; D) Idem for the intrusive body M1. References in Table 1.

5.2. Campanario formation

All samples exhibited good magnetic stability, and each magnetic component was determined with high precision (MAD < 7°) (Fig. 8). Samples from M2 site were characterized by unblocking temperatures between 560 and 590 °C. Two components were obtained: component "a" with unblocking temperatures below 560 °C, and component "b" with temperatures higher than 560 °C (for simplicity these components are labelled "a" and "b" but have no relation to those identified in the Puncoviscana Formation). Samples at M3 presented similar magnetic behavior in all specimens analyzed, observing unblocking temperatures between 590 and 615 °C. Two magnetic components were isolated in almost all specimens: Component "a" was determined from temperatures below 450 °C, while component "b" was determinated with temperatures above 450 °C in most cases. Samples from site M4 behaved similarly to those at M3 with unblocking temperatures of around 590–615 °C. Component "a", however, showed much lower unblocking temperatures (<200 °C).

Mean *in situ* direction for component "a" from sites M2, M3 and M4 is Dec: 356.3°; Inc:−44.0°; n:27; k: 16.4; α95: 7.1°, while after bedding correction is Dec: 24.5°; Inc: −40.7°; n: 27; k: 15.6; α95: 7.3°. Mean *in situ* direction for component b in the M3 and M4 sites is Dec: 65.5°; Inc: −0.2°; n: 38; K: 35.4; α95: 4.0°, while it is Dec:

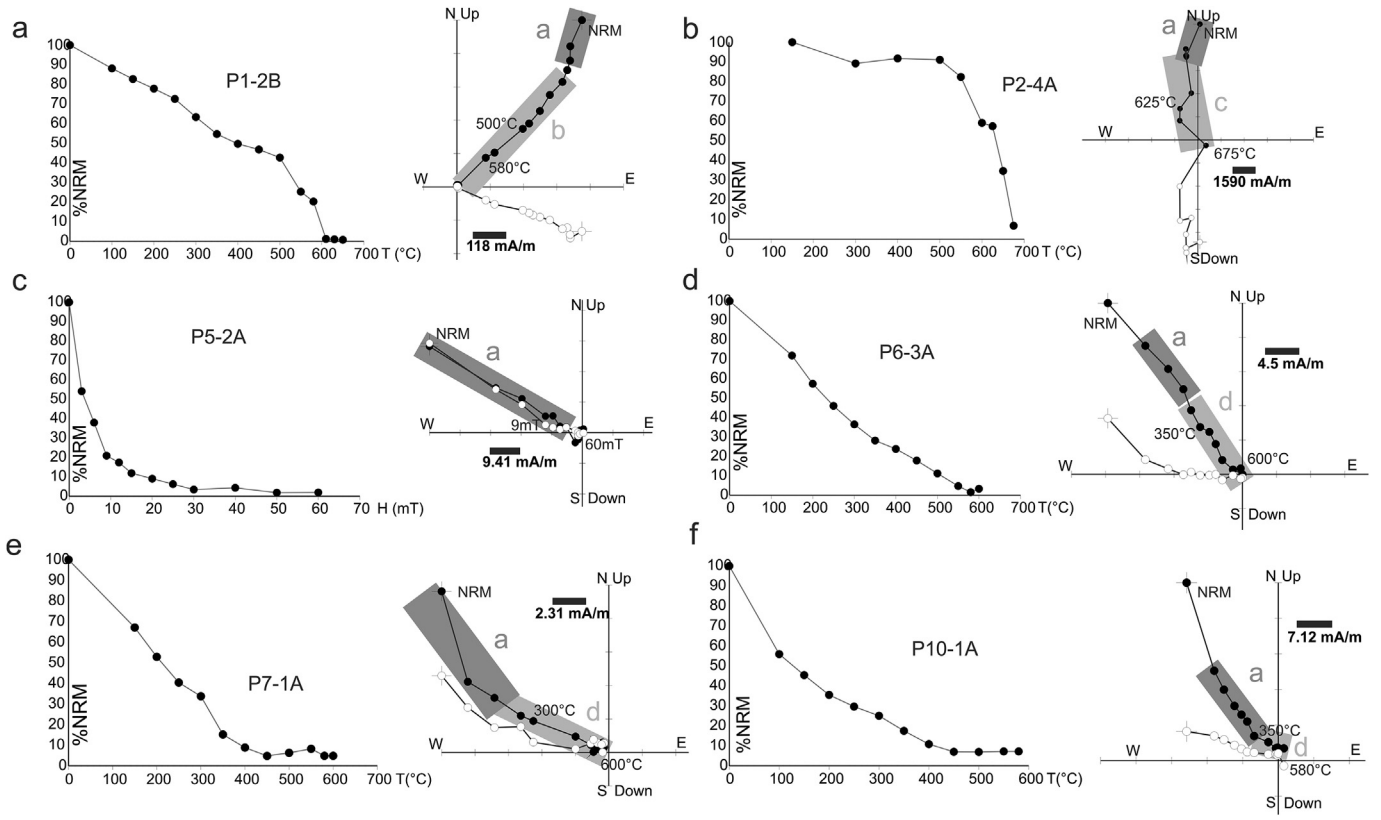


Fig. 5. Diagrams illustrating characteristic magnetic behavior of specimens of the Puncoviscana Formation after bedding correction, demagnetized by alternating fields or high temperatures. Open (solid) circles in Zijderveld diagrams are projections in the vertical (horizontal plane); a) and b) correspond to sites from which VGPs were calculated; c), d), e), f) sites where a recent magnetization was found. Component "a" is illustrated with dark grey shadow and components "b", "c" and "d" with light grey shadow. More references in the text.

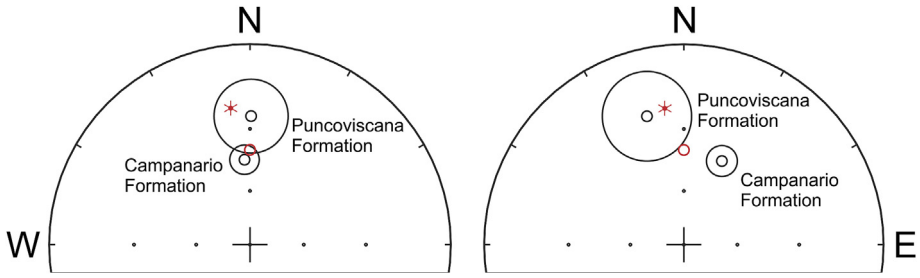


Fig. 6. Mean directions obtained for component "a"; a) *in situ* directions for Puncoviscana and Campanario Formations and the intrusive body (M1); b) idem a) after bedding correction. Red circle, the direction of the axial dipole field at the study locality; red cross, present geomagnetic field at the study area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

62.6°; Inc: 23.0°; k: 27.9; α95: 4.5° after bedding correction. M2 samples did not contribute to the Paleomagnetic pole since they do not carry a high temperature component.

Cores from the Campanario Formation were distributed along 0.35 m of stratigraphic thickness at site M2, 3 m at M3 and other 3 m at M4. Sedimentation rates for the Campanario Formation are not known, however typical values for this kind of sediments is at most of a few hundred mm per thousand years (Seibold and Berger, 1993). Considering that we sampled a stratigraphic interval exceeding 30 m and that no corrections for compaction has been made, a period in the order of 10^5 years is likely represented in our sampling. Therefore we can consider that most, if not all, paleosecular variation must have been averaged-out in the characteristic remanence data of the Campanario Formation.

5.3. Intrusive body M1

The intrusive body sampled at the M1 site presented in most of their specimens a univectorial magnetic behavior (Fig. 8a) and maximum unblocking temperatures near 590 °C. The average *in situ* direction is: Dec: 334.2°; Inc: −50.3°; n: 13; k: 211.7; α95: 2.9°, and Dec: 22.8°; Inc: −44.8° when corrected to the paleohorizontal according to the bedding attitude of the Lizoite Fm beds in which it is intruded.

6. Correction for inclination shallowing

Remanence directions may be affected by compaction and deformation which tend to produce a bias in the inclination towards lesser values. AMS results suggest a depositional to

Table 2
Characteristic remanence directions for the Puncoviscana and Campanario Formations before and after bedding correction. Maximum Angle Deviation (MAD, Kirschvink, 1980) of each specimen direction is presented. In italics and between brackets the values of inclination and declination obtained after correction for inclination flattening to P2 and M3-M4. More references in the text.

Unit	Site	BP	Sample	In situ					Bedding corrected					
				Dec (°)	Inc (°)	MAD/α95 (°)	Lat (°)	Long (°)	Dec (°)	Inc (°)	Lat (°)	Long (°)	α95 (°)	A95 (dp/dm)
Puncoviscana	P1	142/52	1-1A	41.8	−38.4	5.5	51.4	212.1	43.8	13.0	38.3	356.1		
	22°18′11.6″		1-1B	43.5	−35.8	3.2	−49.5	209.7	44.9	15.8	36.6	355.4		
	65°03′49.8″		1-2A	34.1	−30.8	9.1	−57.4	201.0	35.8	19.2	43.0	345.5		
			1-2B	45.6	−36.7	2.8	−47.7	211.2	46.7	15.1	35.3	357.0		
			1-3A	45.7	−43.2	7.9	−48.3	218.2	47.4	8.6	36.6	1.0		
			1-4A	54.8	−37.8	11.1	−39.4	214.6	54.3	14.1	29.2	2.3		
			1-5A	38.6	−30.4	10.6	−53.2	202.6	39.7	20.5	39.1	348.9		
			1-6B	32.2	−37.5	7.4	−60.1	208.5	36.0	12.4	44.6	350.1		
			1-8A	36.2	−34.2	10.0	−56.0	205.7	38.4	16.3	41.5	350.0		
			1-9A	31.5	−32.8	10.3	−60.1	202.1	34.1	16.7	44.5	346.0		
			1-10A	27.0	−51.9	8.7	−64.0	233.9	58.7	5.0	27.6	9.4		
			1-10B	61.8	−46.5	8.4	−34.3	224.2	36.8	−2.5	−48.5	179.6		
			All (n:12)	**40.9**	**−38.4**	**5.4**	**−52.2**	**211.9**	**43.1**	**13.0**	**38.9**	**355.6**		**(3.8/6.4)**
	P2	175/39	2-1A	19.5	51.9	5.4	32.1	314.3	327.2	50.6 (*56.3*)	23.2	266.8		
	22°18′13.9″		2-1B	13.4	57.9	5.4	27.9	306.7	316.5	49.8 (*55.5*)	18.6	258.9		
	65°03′53.2″		2-2A	1.2	51.1	3.9	35.9	296.2	320.0	40.3 (*46.2*)	26.9	255.2		
			2-3A	30.0	47.7	6.2	31.3	325.8	338.3	54.9 (*60.3*)	23.3	277.3		
			2-3B	16.7	50.4	7.9	34.2	312.2	328.0	48.3 (*54.1*)	25.5	266.0		
			2-4A	34.3	42.8	8.4	32.3	332.1	348.4	55.1 (*60.4*)	25.4	285.3		
			2-5B	38.4	46.8	11.8	27.5	333.1	344.4	59.6 (*64.5*)	19.9	283.5		
			2-6A	34.7	50.8	9.2	26.8	327.8	335.2	58.9 (*63.9*)	18.5	276.9		
			2-7A	28.8	45.5	6.1	33.4	325.9	340.9	53.1 (*58.6*)	25.8	278.6		
			2-8A	45.8	43.4	9.9	24.9	340.5	355.7	62.7 (*67.2*)	17.7	292.0		
			2-9A	55.2	52.5	6.6	13.6	340.0	338.7	71.6 (*74.9*)	4.3	284.9		
			All (n:11)	**29.5**	**50.1**	**6.1**	**29.8**	**324.0**	**334.3**	**55.5 (*60.7*)**	**21.7**	**274.4**		**(3.4/6.4)**
	P5	339/37	5-3C	26.6	−0.9	9.1	−56.1	168.3	19.7	−27.2	−69.7	185.4		
	22°17′59.4″		5-4A	353.5	−42.7	12.4	−83.6	1.1	318.9	−40.7	−52.2	15.2		
	65°03′20″		5-6B	11.1	−6.8	14.4	−68.3	146.2	1.6	−24.3	−80.3	124.2		
			5-7A	338.6	−16.9	6.9	−65.3	55.1	328.3	−13.2	−55.7	47.1		
			5-8A	337.4	−48.0	14.4	−68.6	1.9	304.3	−35.6	−38.2	16.8		
			All (n: 5)	**359.3**	**−24.2**	**28.7**	**−80.3**	**110.8**	**339.4**	**−31.2**	**−69.9**	**36.9**		**(42.4/49.3)**
	P6	339/37	6-3A	327.1	−13.4	6.6	−54.7	45.9	321.5	−3.7	−47.4	48.1		
	9 m to East from P5		6-4B	319.7	−28.0	5.5	−51.2	28.7	307.3	−11.5	−36.6	34.6		
			All (n: 2)	**323.6**	**−20.7**	**35.8**	**−53.3**	**37.3**	**314.5**	**−7.7**	**−42.3**	**40.8**		**(27.1/44.1)**
	P7	339/37	7-1A	307.5	−35.6	6.6	−41.2	17.7	294.1	−12.1	−24.6	28.4		
	15 cm up stratigraphically		7-2A	16.4	−22.8	11.7	−71.2	173.9	355.2	−40.3	−85.5	15.3		
			7-2B	308.9	−25.7	10.3	−40.8	26.1	300.4	−4.3	−28.8	35.2		
			All (n: 3)	**330.8**	**−32.2**	**58.9**	**−62.2**	**29.6**	**312.5**	**−20.7**	**−43.2**	**31.2**		**(45.4/73.1)**
	P10	3/57	P10-1A	22.2	−19.9	5.0	−65.6	179.0	355.5	−26.4	−80.6	87.1		
	22°17′58.1″		P10-4B	6.1	−30.1	8.8	−81.6	159.1	338.4	−18.3	−65.6	53.4		
	65°03′18.0″		P10-5A	352.4	−56.6	8.3	−73.7	317.0	308.6	−21.7	−39.8	28.8		
			P10-7A	44.4	−40.9	13.7	−49.2	215.4	337.0	−50.9	−67.5	355.5		
			All (n: 4)	**17.8**	**−38.2**	**26.7**	**−73.5**	**205.4**	**334.6**	**−30.4**	**−65.3**	**34.3**		**(35/43.3)**
	P5-P6-P7-P10		**All (n: 14)**	**352.7**	**−31.0**	**14.2**	**−81.1**	**62.7**	**328.1**	**−25.7**	**−58.5**	**35.6**		**(15.6/21.1)**
			All (N: 4)	**346.8**	**−30.6**	**27.0**	**−76.3**	**47.7**	**324.6**	**−23.0**	**−54.7**	**35.9**	**19.2**	**(10.8/20.4)**
Mesón	M3	189/26	3-1A	59.9	−9.5	1.5	−29.6	197.5	59.7	10.8 (*13.6*)	24.7	5.6		
	22°18′36.8″		3-2A	56.9	12.1	1.6	27.5	4.8	49.5	30.4 (*36.6*)	25.6	347.1		
	65°05′24.0″		3-3A	66.8	18.9	1.1	17.2	6.4	57.9	40 (*46.7*)	14.9	345.6		
			3-4A	69.7	1.2	5.9	18.5	16.3	66.7	23.7 (*29.1*)	14.5	1.0		
			3-4B	68.2	7.0	1.4	18.6	12.9	63.5	28.9 (*34.9*)	15.3	356.1		
			3-5A	61.6	6.2	1.7	24.8	10.2	56.7	26.4 (*32.1*)	21.8	354.1		
			3-6A	64.7	12.3	1.0	20.6	8.7	58.0	33.1 (*39.5*)	18.0	350.4		
			3-7A	62.4	5.8	1.5	24.1	10.8	57.6	26.1 (*31.8*)	21.2	354.8		
			3-7B	68.2	6.4	5.3	18.8	13.2	63.6	28.3 (*34.3*)	15.5	356.5		
			3-8A	59.1	11.4	2.0	25.8	6.3	52.2	30.5 (*36.7*)	23.5	348.8		
			3-9A	79.2	3.1	2.3	9.4	19.3	76.6	27.4 (*33.2*)	4.9	3.0		
			3-10A	53.0	−10.8	4.1	−36.2	194.9	53.5	7.5 (*9.5*)	31.1	4.3		
			3-11A	66.3	−13.7	7.7	−24.5	202.4	67.0	8.3 (*10.5*)	19.0	10.7		
			3-11B	66.3	−10.5	4.8	−23.9	200.7	66.2	11.5 (*14.4*)	18.8	8.3		
			3-12A	65.7	5.5	4.9	21.2	12.5	61.2	26.7 (*32.5*)	18.0	356.3		
			3-13A	78.5	−12.0	3.1	−12.9	206.2	78.5	12.4 (*15.5*)	7.5	13.1		
			3-14A	82.5	5.4	2.7	5.9	19.5	79.9	30.2 (*36.4*)	1.2	2.4		
			3-15A	68.8	−6.3	2.4	−20.8	199.6	67.6	16.2 (*20.2*)	16.2	6.1		
			3-16A	71.3	3.6	2.6	16.5	15.9	67.8	26.3 (*32.0*)	12.7	359.9		
			3-17A	65.3	−9.3	3.5	−24.6	199.7	64.8	12.4 (*15.5*)	19.8	7.1		
			3-18A	65.7	−13.4	1.9	−25.0	202.0	66.3	8.5 (*10.7*)	19.5	10.2		
			3-18B	62.9	−0.1	5.1	−24.9	194.0	59.9	20.7 (*25.5*)	21.3	359.5		
			3-19A	63.4	−17.3	4.2	−27.9	203.2	65.1	4.2 (*5.3*)	21.8	12.3		
			3-20A	71.0	−11.8	3.7	−19.8	203.3	71.0	11.3 (*14.2*)	14.6	10.6		
			3-20B	72.7	−12.8	4.3	−18.4	204.4	72.9	10.6 (*13.3*)	13.0	11.9		

Table 2 (*continued*)

Unit	Site	BP	Sample	In situ					Bedding corrected					
				Dec (°)	Inc (°)	MAD/α95 (°)	Lat (°)	Long (°)	Dec (°)	Inc (°)	Lat (°)	Long (°)	α95 (°)	A95 (dp/dm)
			3-21A	72.8	−7.5	2.8	−17.3	201.8	72.0	15.8 (*19.7*)	12.4	8.3		
			3-22A	64.3	1.9	2.5	23.3	13.6	60.9	22.9 (*28.1*)	19.7	358.6		
			3-23A	67.4	−6.6	1.5	−22.1	199.2	66.3	15.6 (*19.4*)	17.5	5.9		
			3-25A	58.2	−10.5	2.1	−31.4	197.2	58.4	9.4 (*11.8*)	26.3	5.8		
			3-26A	57.9	−1.0	1.5	−29.7	192.0	55.2	18.4 (*22.8*)	26.0	358.3		
			3-26B	61.0	−4.5	3.2	−27.6	195.3	59.3	15.9 (*19.8*)	23.4	2.2		
			3-27A	58.7	−7.2	2.2	−30.2	195.7	57.9	12.7 (*15.9*)	26.4	2.9		
			All (n: 32)	**65.9**	**−2.0**	**3.6**	**−22.6**	**196.3**	**63.6**	**19.6 (*24.2*)**	**18.5**	**1.9**	**3.6**	**3.0**
	M4	174/35	4-1A	65.5	10.9	5.9	20.2	9.7	58.5	43.5 (*50.2*)	12.7	343.4		
	25 m to west from M3		4-1B	55.3	18.9	1.7	27.1	0.4	41.4	47.8 (*54.4*)	20.6	330.3		
			4-2A	46.5	16.0	1.3	35.2	356.4	32.6	41.6 (*48.3*)	29.6	327.6		
			4-3A	60.2	7.1	1.1	25.8	9.1	53.3	38.5 (*45.2*)	18.9	344.1		
			4-4A	59.6	21.6	1.4	22.7	1.4	45.5	52 (*58.3*)	15.4	330.0		
			4-6A	90.0	−18.8	6.5	−3.7	213.8	89.9	16 (*19.9*)	−3.8	15.3		
			All (n:6)	**62.7**	**9.7**	**17.4**	**23.0**	**9.0**	**55.4**	**41.7 (*44.2*)**	**16.2**	**342.0**	**17.4**	**17.4**
	M3-M4		**All (n: 38)**	**65.5**	**−0.2**	**4.0**	**−22.6**	**195.2**	**62.6**	**23 (*27.6*)**	**18.2**	**358.8**	**4.5**	**3.9**

Bold values are the mean directions for each site.

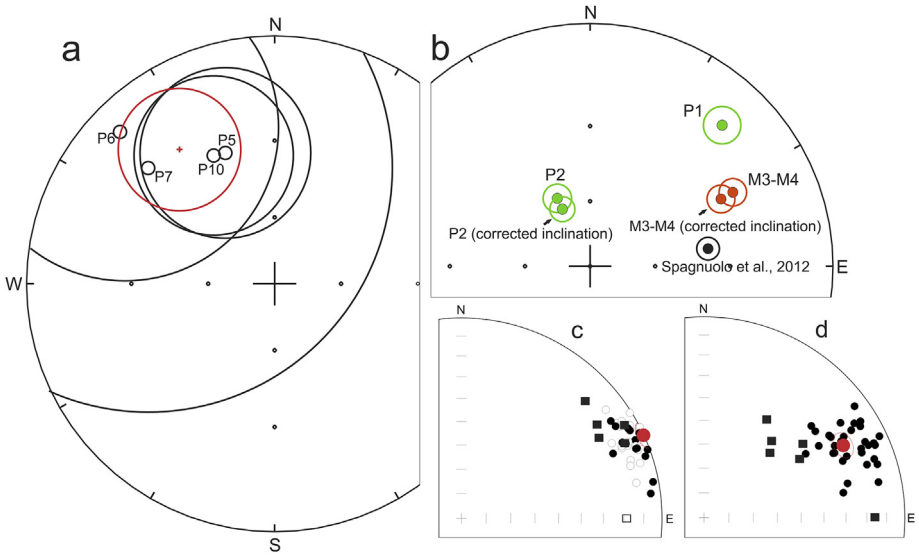


Fig. 7. Stereographic projections of components "b, c and d"; a) mean remanence directions for P5, P6, P7 and P10 after bedding correction. In red, the mean direction (Dec: 328.1°; Inc: −25.7°; n: 14; k: 8.8; a95: 14.2°); b) overall mean sample characteristic remanence directions obtained for the Campanario Formation (sites M3 and M4), before and after correction for inclination flattening (in red) and mean directions obtained for sites P1 and P2 of the Puncoviscana Formation (in green), the latter has also been corrected for inclination flattening (see text for more references); c) specimen characteristic remanent directions *in situ* for sites M3 and M4 of the Campanario Formation; d) idem after structural correction (mean direction in red). Open (full) symbols indicate upward (downward) directions. M2 samples did not carry a high temperature (characteristic) component. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

compactional fabric in the Campanario Fm, while moderate anisotropy degrees in several sites at the Puncoviscana Fm plus microscopic observations indicate internal deformation and low-grade metamorphism. Under these circumstances inclination shallowing of the remanence directions cannot be ruled out. An experimental study was conducted to determine the existence of inclination shallowing due to compaction and/or tectonic deformation at sites P1, P2, M3 and M4. The technique used (Hodych and Buchan, 1994) involves the acquisition of isothermal remanent magnetization (IRM) directed at ca. 45° respect to the bedding plane. If a systematic inclination shallowing is affecting the rocks under study the acquired isothermal remanence would yield lower values of inclination than the applied field and a correction factor can be calculated (Table 3). Previous studies by Spagnuolo et al. (2008) showed no significant inclination shallowing in the Campanario Formation.

Results from this experiment are shown in Table 3. As expected, due to lack of internal deformation, P1 shows no significant inclination shallowing (F = 0.99). However, P2, which shows evidence of internal deformation yielded a mean F value of 1.23. When this factor is applied to the mean inclination of the characteristic remanence at this site the mean inclination changes from 55.5° to 60.7°. For the Campanario formation different parameters were obtained for sites M3 (1.26) and M4 (1.05). The inclination of the mean direction changed after applying the correction from 23.0° to 27.6°.

7. Rock magnetism

To better characterize the remanence carrying minerals, IRM acquisition and backfield curves were produced. Results are shown in Fig. 9a, b. The Puncoviscana formation sites show saturation of

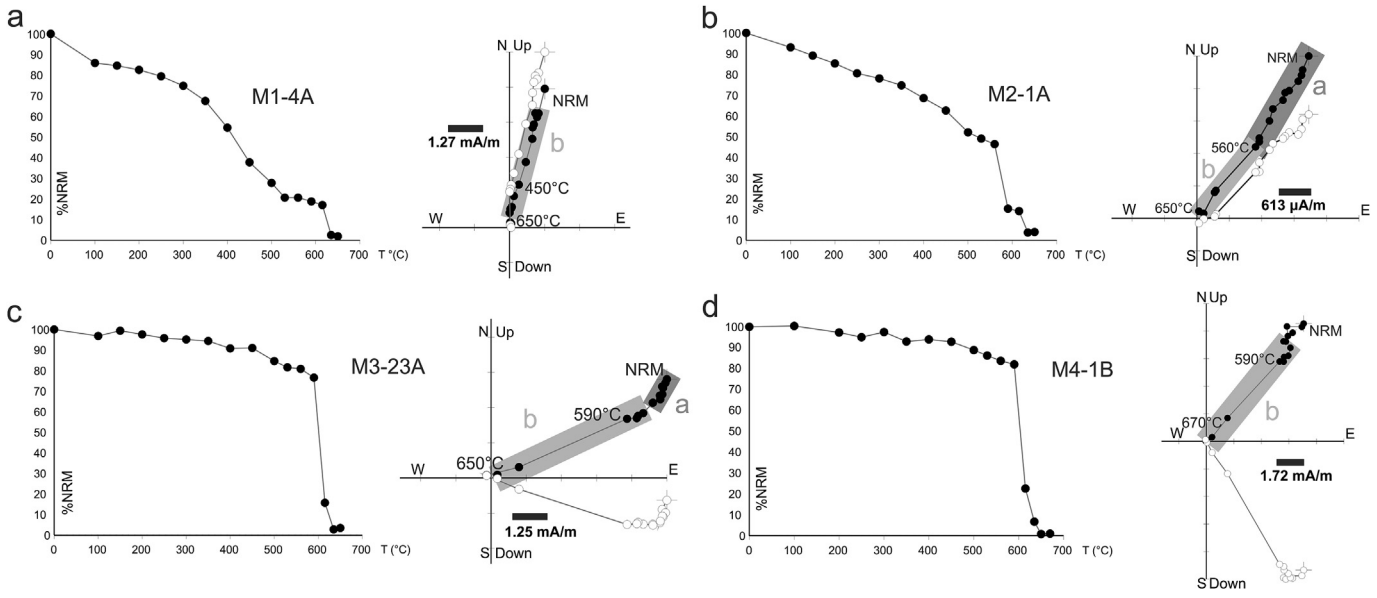


Fig. 8. Zijderveld plots and demagnetization diagrams for representative specimens of a) the intrusive body and b), c) and d), the Campanario formation, after bedding correction. Filled (open) circles in Zijdervled diagrams are projections in the vertical (horizontal plane). Component "a" is represented with dark grey shadow and component "b" with light grey shadow.

Table 3
Results obtained from the oriented isothermal remanent magnetization (IRM) experiment. The inclination is relative to the bedding plane. The IRM was applied in successive steps of 29, 61, 120, 300, 600 and 1000 mT. Steps: indicates the range of values used for computing parameter F (correction factor F, Hodych and Buchan, 1994), n: number of steps, α95: Fisherian statistic mean confidence angle.

Unit	Sample	Applied field Inc (°)	IRM Inc (°)	Steps (mT)	α95 (°)	n	F
Puncoviscana Formation	P1-3A	47.4	47.6	61–1000	1.6	5	0.990
	P1-4A	40.2	41.1	61–1000	0.5	5	0.969
	P1-5A	49.2	48.9	61–1000	2.0	4	1.011
	P1-7A	51.3	51.5	120–1000	1.2	4	0.992
	P2-5A	38.2	31.5	120–1000	1.6	4	1.284
	P2-6B	38.3	34.5	61–1000	1.2	5	1.149
	P2-7B	38.7	31.7	120–1000	0.9	4	1.297
	P2-8C	38.9	34.6	61–1000	0.6	5	1.170
	P2-10C	47.6	41.2	61–1000	1.2	5	1.251
Campanario Formation	M3-6B	51.8	45.4	300–1000	1.5	3	1.253
	M3-16B	52.1	45.2	300–1000	2.0	3	1.276
	M4-2B	60.4	56.7	120–1000	2.5	4	1.156
	M4-3B	71.0	72.2	300–1000	0.8	3	0.932
	M4-5B	53.2	51.8	61–1000	1.0	5	1.052
	M4-6B	51.1	49.2	120–1000	1.6	3	1.070

Mean F: site P1 = 0.99, site P2 = 1.23, site M3 = 1.26, site M4 = 1.05.

magnetization reached well under 1 T. P2 site is the only exception with saturation reached between 1.5 and 2 T. This behavior suggests that a ferrimagnetic fraction (magnetite?) is dominant in the Puncoviscana Formation, although an antiferromagnetic fraction is also present at site P2. On the other hand, the Campanario formation shows a typical antiferromagnetic dominant signal with saturation over 2 T. Similar behavior is shown by the volcanic sill (M1). Unblocking temperatures, as shown previously, suggest that while magnetite dominates in the Puncoviscana Formation, with the only exception of P2, where hematite is dominant, the latter is the carrier of the remanence in the Campanario Formation and the volcanic sill.

IRM back-field curves were made on samples of the Puncoviscana formation to obtain remanence coercivity (Hcr) values. As shown in Fig. 9d there is a clear difference between P1 and P2 sites with Hcr values of 46.9 and 76.4 mT, respectively. P5, P6, P7 and P10 sites show lower Hcr, ranging from 23.4 to 36.2 mT. A correlation between higher Hcr and directional consistency is observed, suggesting that sites with lower Hcr values are likely dominated by multidomain magnetic fractions and therefore more unstable magnetic carriers. Fig. 9e illustrates the existence of large (>200 μm) secondary magnetite crystals at site P10.

Low temperature thermo-magnetic (k vs T) curves showed Verwey transitions to ~ −155 °C in samples from sites P1, P2, P5, P6, P7 and P10 (Fig. 10a–c), which is indicative of multidomain magnetite (Dunlop and Ozdemir, 1997). In these sites there are near-reversible cycles in high temperature curves with well defined Hopkinson peaks between 550 and 580 °C also suggesting the existence of SD magnetite (Dunlop and Ozdemir, 1997) or small PSD grains (Dunlop, 2014), in particular at site P1. This suggests a combination of SD and MD magnetite grains at these sites with higher proportion of MD grains at those sites with lower Hcr and more unstable remanence behavior during demagnetization (P5, P6, P7 and P10). High temperature curves show the apparent

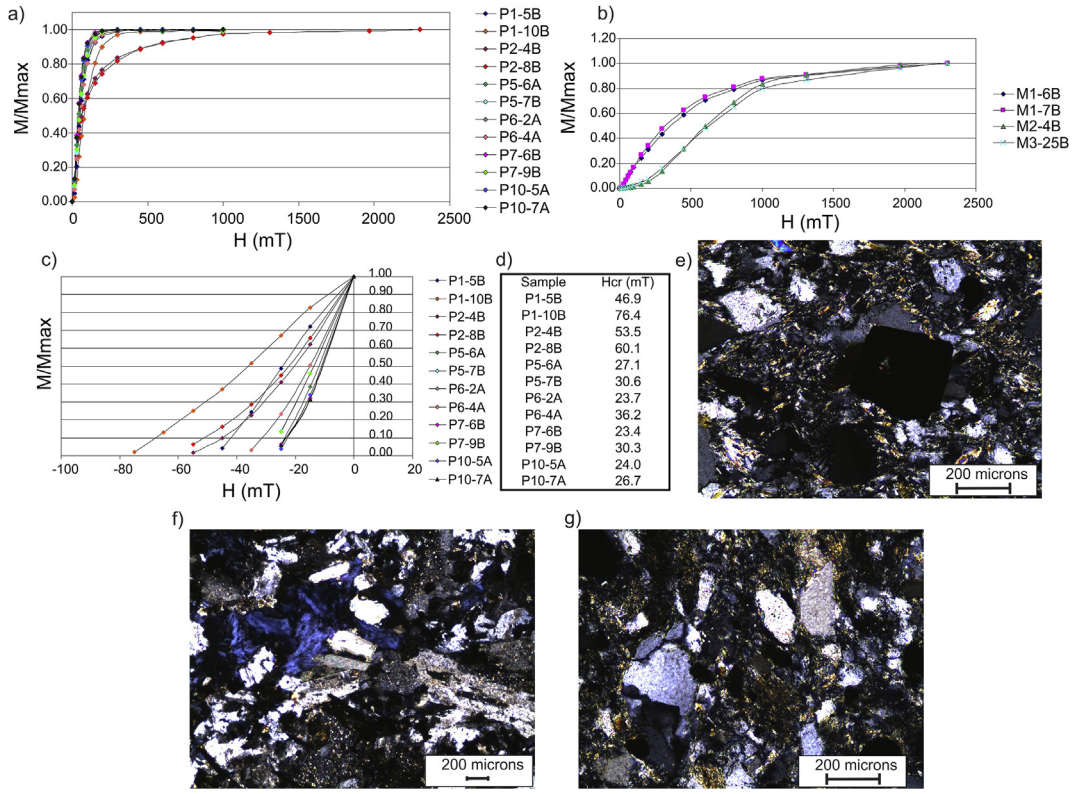


Fig. 9. Normalized acquisition curves of isothermal remanent magnetization (IRM); a) curves belonging to Puncoviscana Formation sites; b) idem for the Campanario Formation and Cenozoic sill; c) back-field magnetization curves for the Puncoviscana Formation; d) values obtained for remanence coercitivity (Hcr) in the study samples; e) petrographic thin section of a specimen from site P10; f) and g) idem for sites P1 and P2, respectively. Note the size of the opaque crystals of magnetite.

presence of a contributor to the bulk susceptibility over 580 °C, suggesting the existence of hematite. This is consistent with NRM demagnetization and IRM acquisition curves for P2 (Figs. 5 and 9). However, this is not suggested by these experimental results in the case of P1, P5, P6, P7 and P10. Samples from sites M1, M2, M3 and M4 showed irreversible curves with mineralogical changes after 400 °C, indicating the generation of high susceptibility minerals due to experimental heating. Formation of magnetite is suggested from the cooling curve, probably from paramagnetic clay minerals.

Bulk susceptibility measurements made during stepwise high temperature demagnetization showed no systematic changes up to around 550–600 °C. The linear decay of the natural remanent magnetization during thermal cleaning, however, suggests that if new minerals were formed during the experimental procedures, they did not carry significant remanence.

8. Discussion

The scarce paleomagnetic information on Neoproterozoic to Early Paleozoic exposed units in northwestern Argentina severely limits the knowledge of the paleogeographic evolution of the Pampia terrane and its relations to other Gondwana blocks during the assembly of this supercontinent. Spagnuolo et al. (2012), based on two paleomagnetic poles obtained in the Campanario formation and the overlying Santa Rosita formation, proposed a dextral displacement of Pampia along the margin of the Rio de la Plata craton from the present southern edge of the Kalahari craton along a strike-slip or transform fault system to a present-day position during the Cambrian. Such displacement was proposed on the base of the inconsistency between the paleomagnetic poles obtained from the Campanario Formation (Pampia terrane) and the Early Paleozoic reference poles for Gondwana. Spagnuolo et al. (2012) obtained their paleomagnetic data from two localities over 30 km apart (Fig. 11). Our study on the same formation was located over 60 km to the north from the previous ones. The new mean direction of the Campanario formation, obtained in Santa Victoria Oeste, even after being corrected by “inclination flattening” differs from that obtained for the same unit in the Matancillas and Iruya localities (Spagnuolo et al., 2008, 2012). The largest discrepancy between the two mean directions is in declination, with a difference of around 20° (see Table 4). This could be interpreted as due to an *in situ* clockwise rotation of the southernmost localities with respect to the northern one. This interpretation was originally preferred by Spagnuolo et al. (2008) with data only from the Matancillas locality. However, it was considered less likely when identical directions were found in the Campanario Formation exposed in Iruya, 30 km WSW from the first location (Spagnuolo et al., 2012). The existence of a smaller discrepancy of the paleomagnetic pole of the earliest Ordovician Santa Rosita formation obtained in the Matancillas locality, respect to the reference pole of Gondwana, reinforced the idea of the improbability of *in situ* Andean rotations to explain the anomaly of the paleomagnetic directions of the Campanario formation. Systematic tectonic rotations in the Central Andes have been described since several decades ago. This has been labelled as “The Central Andean Rotation Pattern” (CARP) by Somoza et al. (1996) and predicts the existence of clockwise rotations to the south and anticlockwise rotations to the north of the Arica bend at around 18°S. However, more recent studies (eg. Arriagada et al., 2006) indicate that the pattern of rotations is more complex. The existence of tectonic deformation transfer zones in the Andean orogenic front (Kley, 1996; Kley and Monaldi, 1999) may produce sectors of anomalously large *in situ* rotations (see discussion in

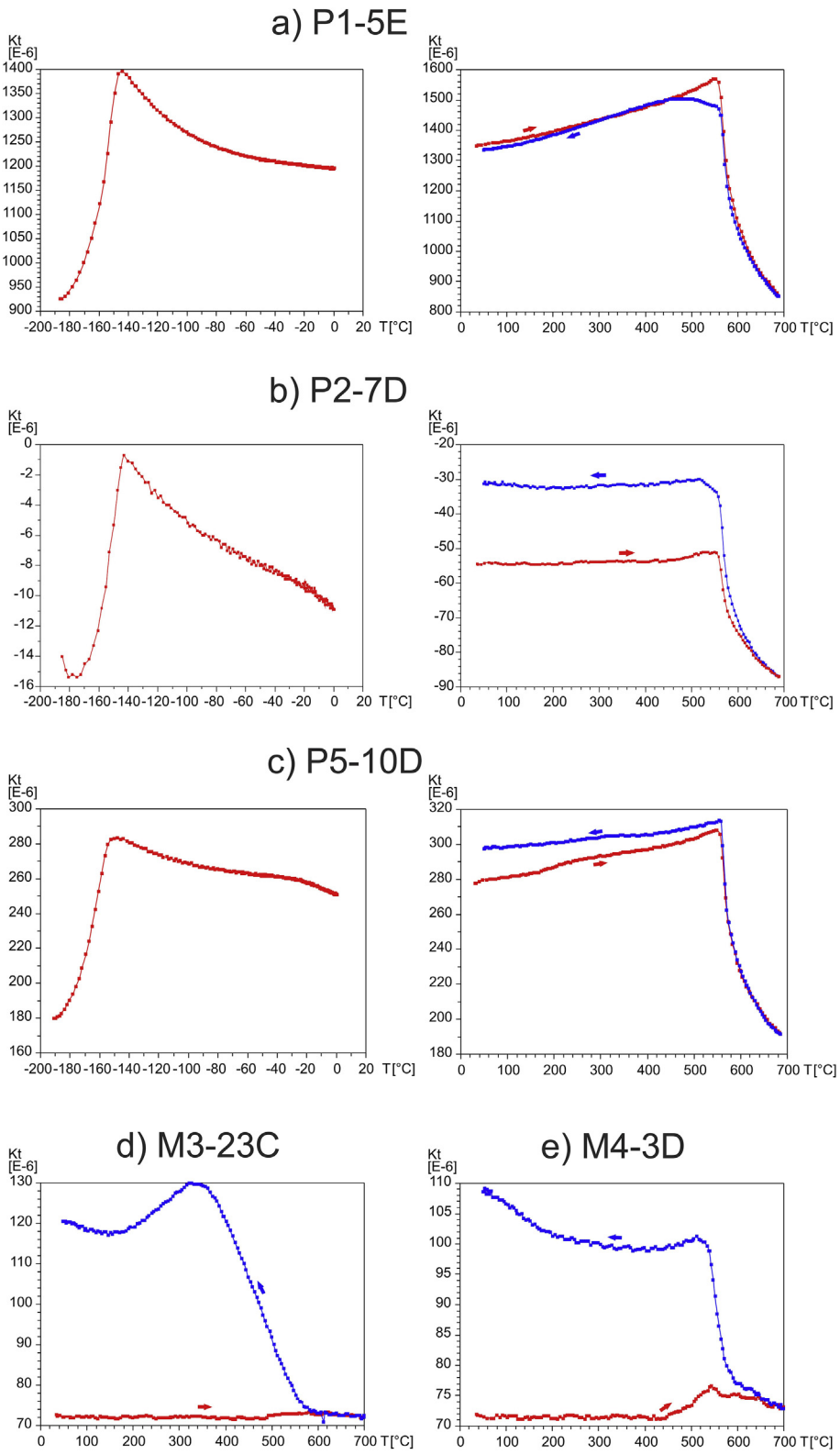


Fig. 10. Representative low and high temperature thermomagnetic curves (k vs T) for the Puncoviscana and Campanario Formations. a) P1, b) P2, c) P5 and d) M3 and M4. Red lines correspond to heating curves, blue to cooling trajectories. Note Verwey transitions (around −160 °C) in the Puncoviscana samples. Small Hopkinson peaks are apparent in the high-temperature curves of the Puncoviscana Formation. The Campanario Formation samples show clear irreversible curves with neoformation of magnetite. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

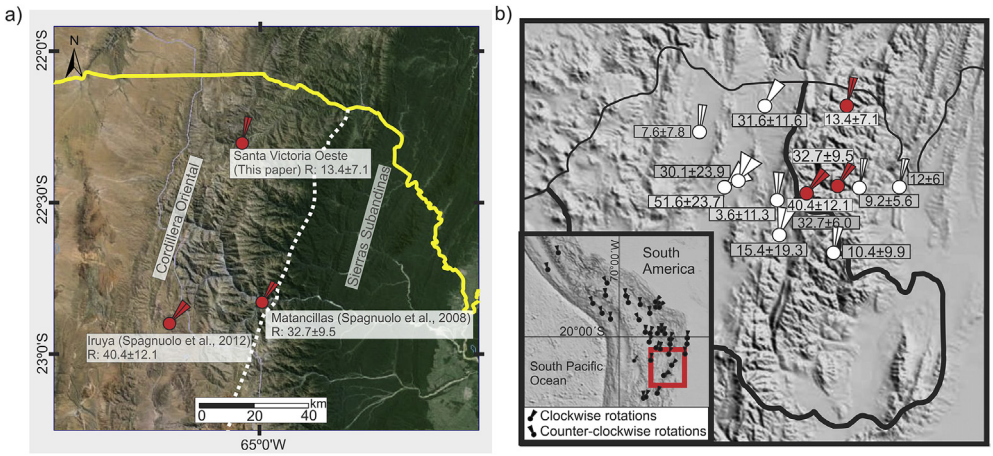


Fig. 11. a) Declination anomalies of the Campanario Formation at the Santa Victoria Oeste (this study), Matancillas (Spagnuolo et al., 2008) and Iruya (Spagnuolo et al., 2012) b) Rotations found for the Campanario Formation (red symbols) compared with rotations for nearby Cenozoic units (white symbols). More references in the text and Table 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
"In situ" rotations computed for the Campanario and Santa Rosita Formations. Gondwana reference poles for 495 Ma (Grunow and Encarnación, 2000) and 485 Ma (McElhinny et al., 2003) were used to compute the declination anomalies for the Campanario and the Santa Rosita Formations, respectively. R: rotation; F: anomaly in inclination ("flattening"); ΔR and ΔF are the errors in rotation and flattening, respectively, computed according to Demarest (1983).

Sampling location	R (°)	ΔR (°)	F (°)	ΔF (°)
Campanario F. (Santa Victoria oeste, this paper)	13.4	7.1	21.9	6.7
Campanario F. (Matancillas, Spagnuolo et al., 2008)	32.7	9.5	5.6	7.8
Campanario F. (Iruya, Spagnuolo et al., 2012)	40.4	12.1	10.3	9.9
Santa Rosita F. (Matancillas and Iruya, Spagnuolo et al., 2012)	20.1	12.8	13.2	10.4

Spagnuolo et al., 2008). The new results on the Campanario Formation could suggest the existence of such a transfer zone at the latitude of the Matancillas and Iruya locations explaining why the Cambrian rocks at these two localities may have undergone significantly larger clockwise rotations than the same unit at Santa Victoria Oeste.

Fig. 11b illustrates the tectonic rotations found in Cenozoic rocks in the study region (taken from Spagnuolo et al., 2012). Note that the larger rotations observed in the Campanario formation in Matancillas and Iruya (Table 4) are not reproduced by Cenozoic units exposed close to these locations (Fig. 11). The declination anomaly computed for the Campanario Formation at Santa Victoria Oeste is of 13.4 ± 7.1°, if the reference poles for Gondwana used (Table 4) are the same than those selected by Spagnuolo et al. (2012). The absence of significant rotations in the neighboring localities suggests that rotation in Iruya and Matancillas must have occurred prior to the deposition of the Neogene sediments of the Cordillera Oriental and Sierras Subandinas (see Echavarría et al., 2003) or consist in complex systems of variable rotations of small blocks.

A complex pattern of Andean rotations (Fig. 11) could explain most of the anomaly in the positions of the Campanario formation paleomagnetic poles, although a small paleolatitudinal anomaly remains (Fig. 12b). This could be ascribed to some difference in the age of the study formation or of the remanence acquisition at different localities, during a relatively fast polar wander of Gondwana, or, incomplete removal of inclination shallowing in Santa Victoria Oeste, despite the experimental procedure performed.

According to Spagnuolo et al. (2012), the model of Andean rotations cannot explain the smaller rotation values found in the Santa Rosita Formation (Early Ordovician) in the localities of Matancillas and Caspalá-Santa Ana and the lack of appreciable rotation of the Devonian sediments of the Baritú Formation in the latter place. Together with the low to non-significant rotation values found in Neogene sediments at nearby localities (see Spagnuolo et al., 2012 and references therein), they pose some problems for this explanation.

Spagnuolo et al. (2012) preferred an alternative interpretation that considers that the paleomagnetic poles of the Campanario and Santa Rosita formations are representative of the Pampia terrane. Our new pole for the Campanario formation, however, differs from the previous ones and weakens this model. Under this assumption, the only possible explanation is to assume that the levels sampled in Santa Victoria Oeste – or their magnetization – are younger than those at the localities of Iruya and Matancillas and that the observed difference is the result of the apparent polar wander of Pampia. At this respect it is interesting to note that U-Pb radiometric determinations on the youngest detrital zircons of the Campanario Formation at the study area are significantly younger (501 Ma, Franceschinis et al., in preparation) than at a southern locality (514 Ma, Augustsson et al., 2011). As shown in Fig. 12a, if Pampia is placed approximately halfway between its final position in Gondwana (present-day relative position in South America) and the proposed Early Cambrian location adjacent to the Kalahari craton, as suggested by Spagnuolo et al. (2012), the paleomagnetic pole obtained in this study becomes consistent with the reference paleomagnetic poles for Gondwana for late Early Cambrian- Early Ordovician (505 Ma, following Spagnuolo, 2009; 495 Ma, Grunow and Encarnación, 2000; 485 Ma, McElhinny et al., 2003) and does not match the reference poles when Pampia is translated up to the Kalahari craton margin ("full displacement"). According to this interpretation, the different paleomagnetic poles of the Campanario formation may be recording the displacement of Pampia along the western margin (present coordinates) of the Rio de la

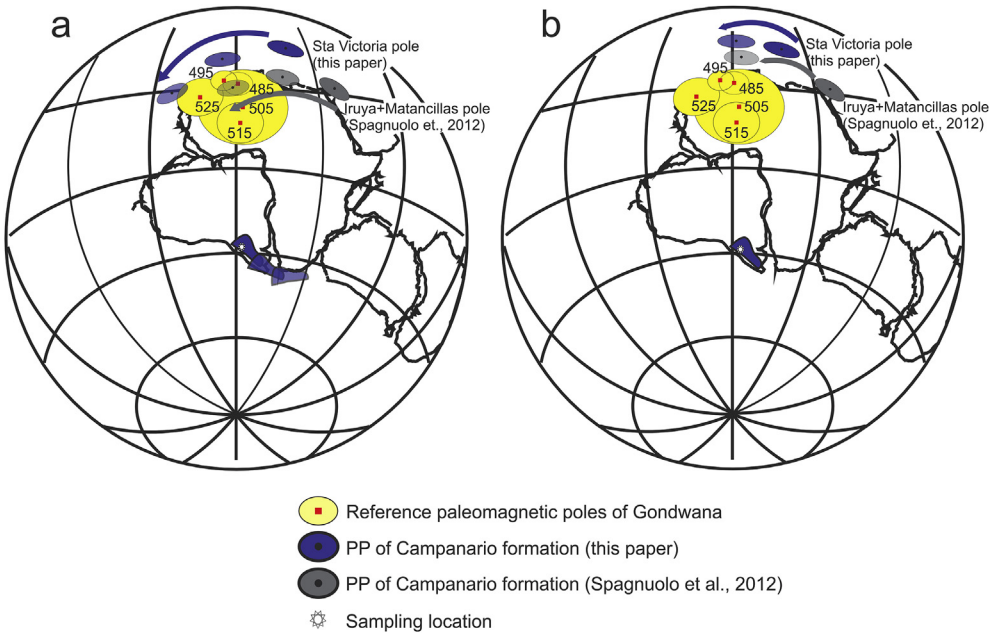


Fig. 12. Paleomagnetic poles of the Campanario Formation rotated to Africa in the Gondwana framework (43.02°N 329.93°E 58.84°; Reeves et al., 2004) compared with the reference paleomagnetic poles of Gondwana for 525, 515, 505 (following Spagnuolo, 2009); 495 (Grunow and Encarnación, 2000); and 485 Ma (McElhinny et al., 2003). Gondwana reconstruction after Reeves et al., 2004 and Trindade et al., 2006. On the left side, paleopoles location with the displacement of Pampia; at right side, paleopoles location with hypothetical Andean rotation. Pampia rotated according to an Euler pole at 10.1°N 19.5°E (full rotation: 46°, half-way rotation: 23° counterclockwise) following Spagnuolo et al. (2012). PP: Paleomagnetic pole.

Plata craton from an original position close to Kalahari (eg Schwartz and Gromet, 2004; Rapela et al., 2007, 2016).

If instead of reference poles for Gondwana, a recent apparent polar wander path (Rapalini et al., 2015) is used for comparison (Fig. 13), we see that a mid-way displacement of Pampia would make the original pole by Spagnuolo et al. (2012) compatible with a ca. 500 Ma position, while a full displacement would turn it consistent with an older position (>510 Ma). Our new pole, however, does not experience any significant improvement in its comparison with that path either with a "mid-way" or a full displacement of Pampia as it falls on a section of the Gondwana APWP that is too old. Possible Andean rotations, on the other hand (Fig. 13d), do not produce a better consistency of the Campanario poles with the Gondwana path. This analysis highlight the importance of a more robust and well dated Early Paleozoic Gondwana apparent polar wander path for more conclusive interpretations of the Pampia paleomagnetic data.

The paleomagnetic information provided by the Puncoviscana formation should be used carefully because we only obtained two VGPs, which do not coincide with each other and probably reflect different ages of magnetization. Fig. 13a shows that there is no overlap between these VGPs and the apparent polar wander path for the Río de la Plata craton and most of Gondwana between about 550 and 500 Ma (Rapalini et al., 2015). P1 is close to the positions of the curve for the late Early Cambrian to the Early Ordovician (525 Ma–475 Ma). The sill sampled at this site presents scarce or null evidence of deformation and/or metamorphism, suggesting that it is younger than the Tilcaric deformational phase that affected the Puncoviscana formation, i.e. < 525 Ma. If Pampia is displaced "mid-way" to the SE (in current coordinates) P1 becomes more consistent with respect to the Gondwana path at the likely age of this intrusive. P2 corresponds to a tuff level interbedded in the Puncoviscana Formation, with a radiometric age of 537 ± 0.9 Ma (Escayola et al., 2011). It's position is relatively close to the 535 Ma part of the Río de la Plata-Gondwana path. Any displacement of Pampia turns the position of P2 closer to sections of the path that are older than 537 Ma, suggesting little or no displacement of Pampia relative to the Rio de la Plata craton since the early Cambrian. The contradictory conclusions arising from P1 and P2 highlight, among other things, the VGP character of P1 and P2, which turns any conclusion from these data alone still speculative. Fig. 13c shows that rotation of these VGPs according to the "Andean rotation scenario" does not produce any significant change in their position respect to the Rio de la Plata-Gondwana path.

9. Conclusions

A new paleomagnetic study was carried out in the early Late Cambrian Campanario Formation and the latest Ediacaran – Early Cambrian Puncoviscana Formation, exposed in the Santa Victoria Oeste locality in northwestern Argentina. Magnetic fabrics of the study rocks were investigated by anisotropy of magnetic susceptibility which indicated a pre-tectonic (pre-Andean) fabric. Stepwise demagnetization and rock magnetic studies indicated that a characteristic, probably primary, remanence carried by hematite is present in the Campanario Formation and permitted to compute a new paleomagnetic pole for this unit (18.2°S, 358.8°W, K: 27.9; A95: 3.9). This new pole does not coincide with a previous one obtained from the same unit at localities 60 km south from Santa Victoria Oeste. This discrepancy is attributed either to large tectonic rotations associated to the Andean orogeny and affecting the southern localities, or a recording of the displacement of Pampia during its accretion to Gondwana. The paleomagnetic study on the Puncoviscana Formation only yielded two virtual geomagnetic poles that do not coincide but are not too far from their expected positions on the Rio de la Plata – Gondwana late Ediacaran – Cambrian apparent polar wander path.

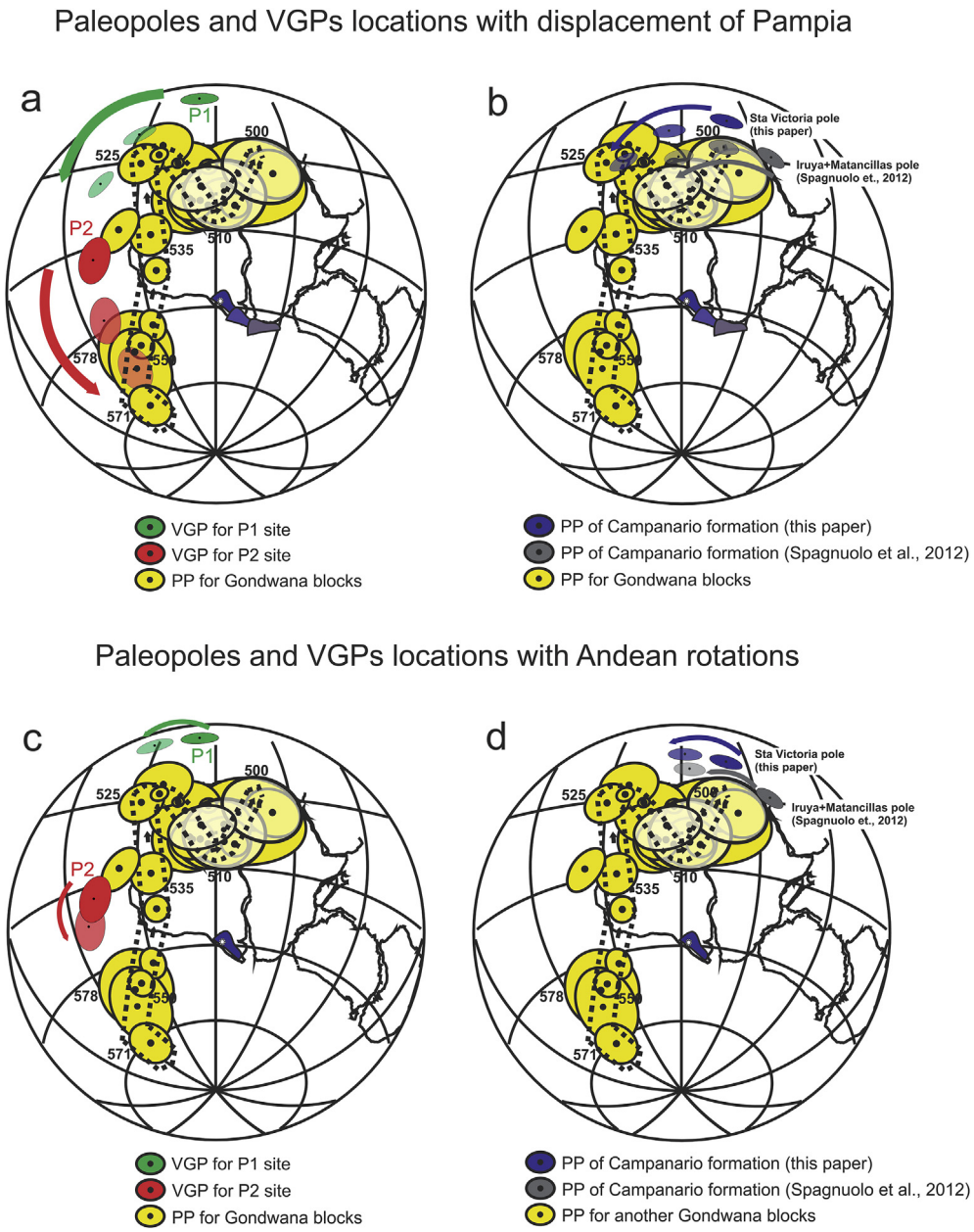


Fig. 13. a) Location of VGPs belonging to the Puncoviscana Formation with respect to the APWP of Gondwana for the Late Neoproterozoic – Early Paleozoic (Rapalini et al., 2015) according to the different locations of Pampia along the Gondwana margin presented in Fig. 12; b) idem for the Campanario Formation paleopoles (in blue, this paper; in grey, Spagnuolo et al., 2012); c) location of VGPs of the Puncoviscana Formation before and after applying a possible Andean rotation affecting the Santa Victoria Oeste locality; d) idem for the paleomagnetic poles of the Campanario Formation. Rotations as in Fig. 12 The white star is the sampling location. More references and discussion in the text. PP: Paleomagnetic pole. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

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